

Empiricist

P0 cards · H-algo/H-size · GPU honesty · directional success bar

Scout success

Documented stage-isolated cards — NOT “we reproduced RoboGen endless skill factory.” Success bar = directional, NOT 0.774 parity.

P0 cards from DELIVERABLE

Proposal · Verify · algo · gap — stage-isolated smokes

P0 Proposal

Fig. 2 A-B

N=10-20 JSON tasks
schema + collision pass
API \$ / CPU only

P0 Verify

Fig. 4

Object $\pm$ size verify
5-7 assets · BLIP-2
0-1 GPU-h + VLM \$

P0 Algo

Fig. 5

MP-hybrid vs RL-lite
1 articulated stub
 $\leq 50-100k$ SAC or MP

P0 Gap

proxy

Train synth $\rightarrow$ eval DR
 Δ SR under appearance
1-8 GPU-h

Hypotheses — H-algo / H-size first

Falsifiable course smokes

H-algo MP-primitive $\pm$ short RL $\gg$ RL-only SR on articulated stubs (Fig. 5 moral)

H-size Dropping size verify hurts BLIP-2 / checklist more than object verify alone (Fig. 4)

H-object Object verify reduces retrieval variance / false accepts

H-proposal Stage A/B stub $\geq 70\%$ schema-valid (paper scene fail $\sim 8\%$; smoke looser)

H-gap Privileged/synth-trained policy drops under appearance/camera DR

H-coverage Generative task text @ matched N has lower Self-BLEU than fixed suite

Verify card — Fig. 4 protocol

Size verify dominates; object verify ↓ variance

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

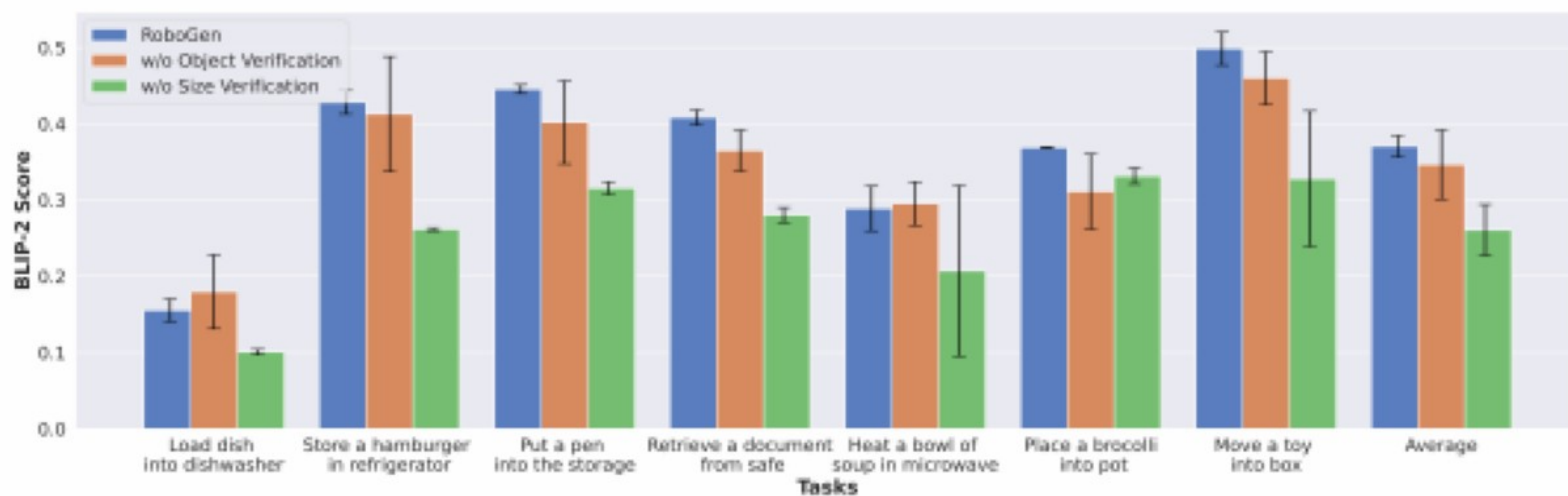


Figure 4: We compare the BLIP-2 score of ablations of RoboGen on 7 tasks to evaluate the importance of both object and size verification.



Algo card — Fig. 5 protocol

Hybrid > RL-lite on ONE articulated open/close · not 12-task suite

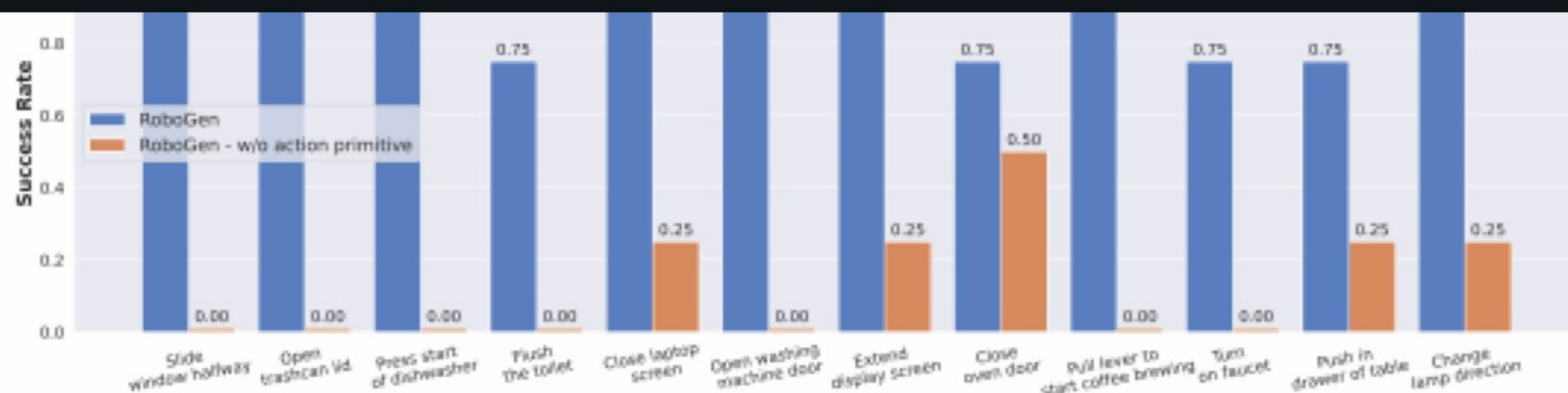


Figure 5: Among 12 articulated object manipulation tasks, the success rate decreases drastically if only RL is used for skill learning.

tion tasks, meanwhile leverage more diverse assets retrieved from open-world databases such as Objaverse, resulting in much higher diversity in both task semantics and scene images. We also provide the full list of generated tasks in

judge whether the joint angle value 0 corresponds to the door being opened or closed. 3) failure to find matched assets for tasks that require extremely precise spatial relationships, as it is hard to retrieve assets that exactly match the

Motion deep dive — Empiricist hooks

Log algorithm chosen · wall-clock · N=8 best-of per subtask

SAC

lr 3e-4
MLP [256]³
1M steps/sub
horiz 100 / fs 2
6D EE

BIT*

OMPL
0.03 m pre-contact
surface-normal grasp
approach/release

Adam

soft traj-opt
300 steps · lr 0.05
EMD cost
horiz 150/200

CEM

loco GT dyn
horiz 150 / fs 4
joint-angle acts
more stable here

Long-horizon + skill structure

Sequential subtasks · N=8 best-state handoff · avg 3.13 substeps · ~1.5 RL + ~1.63 MP · smoke $\leq 3-4$ substeps; defer 8-10

Fig. 6 — measure the hybrid product

Substep / RL / MP histograms · wall-clock morals

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

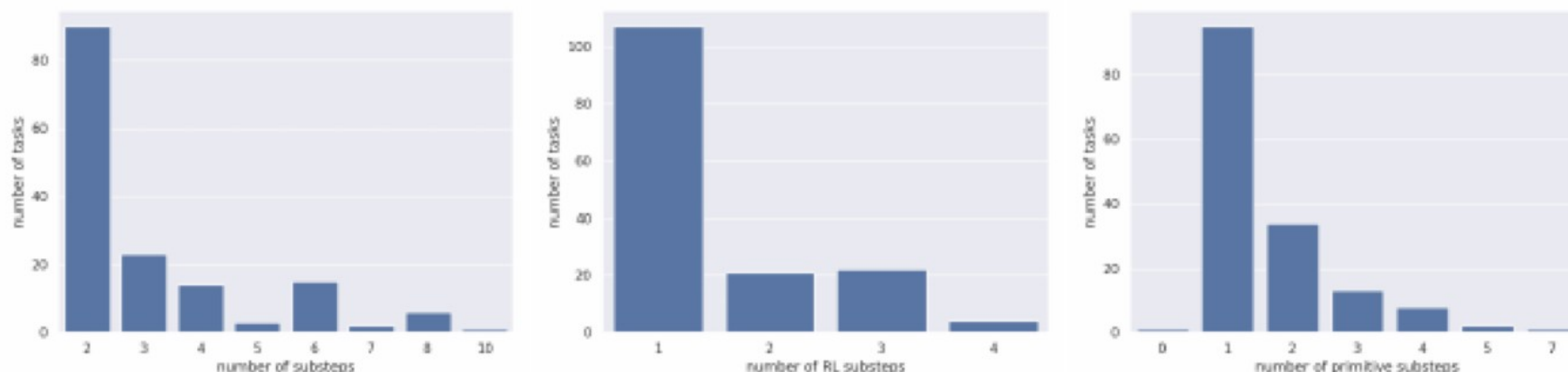


Figure 6: Left: The distribution of number of substeps for the generated rigid and articulated object manipulation tasks in Table 2. The average number of substeps is 3.13. Middle: The distribution of number of substeps that need to be solved using RL for the generated tasks. The average number of RL substeps is 1.5. Right: The distribution of number of substeps that need to be solved using motion planning based primitives for the generated tasks. The average number of such kind of substeps is 1.63. Regarding duration for solving the task: if the task’s subgoals can all be solved via planning, typically each task can be solved within 10 minutes. If certain subgoals require RL to solve, it usually takes around 2-3 hours for each RL-necessary step, and the total duration thus depends on both the number and nature of the subtasks. Taking these into account, a task typically takes 4-5 hours on average. This is done using 8 threads of a CPU running at 2.5Ghz, meaning that each single node in a cluster with a 32-core (64 threads) CPU could run 8 jobs in parallel at the same time.

Hyperparams to pin (App A.3)

Lock these on smoke cards — cut 1M steps for course

SAC

lr $3e-4$ · MLP $[256]^3$ · 1M steps/sub (smoke: cut to $\leq 50-100k$) · horiz 100 / fs 2 · 6D

BIT*

pre-contact 0.03 m · surface-normal sample · push-to-contact · OMPL

Soft Adam

300 grad steps · lr 0.05 · EMD · horiz 150 or 200 · needs diff-sim

CEM loco

horiz 150 · fs 4 · GT dynamics · joint angles

GPT temps

proposal 0.8–1.0 (diversity) · other stages 0–0.3 (robustness)

Retrieve

Sentence-BERT top-k=10 Objaverse · Gemini-Pro caption → GPT-4 verify

GPU / RAM / TIME honesty

Physics sim = CPU-heavy · LLM = API \$ · SAC MLP = modest VRAM

Proposal N=20

0 GPU

10-40 min + API \$

Verify 5-7 assets

0-8 GB

30-90 min

MP-only 1 task

0-4 GB

minutes-1 h

SAC-lite $\leq 100k$

4-12 GB

1-4 h

Paper 1M $\times$ 1.5 RL

4-12 GB

$\sim 2-3$ h $\times$ 1.5 / task (CPU acct)

Full 106 propose $\rightarrow$ learn

mixed

days-weeks · N for course

Success bar — directional NOT 0.774 parity

Pass ≥ 2 P0 cards with stage cost logs

Feasible (pass)

- Proposal stub valid varied @ N=20
- Fig. 4 moral: size ($\pm$ object) $\uparrow$ alignment
- Hybrid $>$ RL-lite on ONE articulated
- Gap proxy shows Δ SR under DR
- Automated proxy SR — not human video

NOT required / overclaim

- Table 1 106-task regen exact scalars
- Human-video 69-task 0.774 average
- Fig. 5 12-task suite reproduction
- Genesis soft / loco Tables 6–7
- “Reproduced RoboGen / beat Beh-100”

Honesty rule

Log stage · LLM model+T · env steps · API tokens/\$ · peak VRAM. Never amortize paper 0.774 onto a Partial stub.

Proposal stub + gap proxy detail

Smoke A + Smoke E from DELIVERABLE

Smoke A — proposal

- Prompt → JSON {task, assets, sizes, init, layout}
- Validate schema; optional collision push
- Metric: schema validity; collision resolve rate
- N=10-20; mock or cheap LLM; no SAC

Smoke E — gap proxy

- Tiny BC on synthetic demos
- Eval under rand textures / camera
- Report ΔSR as sim↔real PROXY
- Paper leaves transfer open (§5)

Defer / N

Genesis soft fleet · 1M×N SAC · Objaverse full · 12-loco CEM · real Franka deploy — out of course smoke.

env outline/ — outlines only

Do NOT scratch from quiet box · public sim stand-in

slurm_proposal_stub.sh

P0 N=20 proposal

slurm_verify_ablation.sh

P0 Fig. 4 verify

slurm_planner_vs_rl.sh

P0 Fig. 5 hybrid

slurm_skill_tiny_synth.sh

P1 tiny skill/BC

slurm_gap_proxy.sh

P0/P1 DR gap

slurm_diversity_card.sh

P1 Self-BLEU small-N

Numbers tattoo — pin these

Every deck · Ablation Bot MISSING (paper extract)

0.284

Self-BLEU (best)

0.165

SentBert ↓

0.193

Scene ViT ↓

0.762

Scene CLIP ↓

0.774

success / 69

19 / 155

validity fails

3.13

avg substeps

Fig. 5

RL-only collapses

0.745

manip SR (n=50)

~0.886

soft SR (n=7)

~0.833

loco SR (n=12)

~1.5+1.63

RL + MP avg

Empiricist — close

Takeaways + exam bite

1. Run P0 verify + algo (+ proposal) with stage cost logs.
2. Pin SAC $1\text{M}/3\text{e-}4/[256]^3$ — cut steps for smoke.
3. GPU honesty: CPU sim + API \$ dominate; never claim 0.774 from stubs.
4. Exam: competence = decomp + choose MP/RL/traj-opt — not E2E LLM policy.

EXAM BITE

Directional H-algo / H-size cards beat slogan reproduction.